ONBRD-10: Building Dashboards

> **Series:** ONBRD — Dynatrace Onboarding | **Notebook:** 10 of 10 | **Created:** December 2025 | **Last Updated:** 10/02/2026

Visualizing Your Data

Dashboards provide at-a-glance visibility into your environment's health and performance. This notebook covers dashboard creation, common visualization patterns, and sharing with your team.

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Prerequisites

- Viewer access or higher
- Familiarity with DQL (ONBRD-08)
- Data in your environment to visualize

1. Dashboards vs Notebooks

Dynatrace offers two visualization tools:

Feature	Dashboard	Notebook
Purpose	Monitoring at a glance	Interactive analysis
Layout	Fixed grid tiles	Sequential document
Auto-refresh	Yes	Manual
Interactivity	Click-through links	Query editing
Sharing	Wall displays, reports	Investigation documentation
Best for	NOC screens, status pages	Troubleshooting, exploration

This notebook uses **Dashboards**, the Latest Dynatrace app. Dashboards Classic is the older surface, and Dynatrace's advice for it is: *"If you're still using classic dashboards, we encourage you to upgrade your dashboards and benefit from all the latest dashboarding possibilities made available by the Dashboards app in the latest Dynatrace."* Classic tiles such as **Host health** and **Service health** do not exist in the Dashboards app; their job is done with DQL tiles (§5).

> **Sources:** Dashboards Classic (DT docs).

When to Use Each

Scenario	Use
Team status display	Dashboard
Incident investigation	Notebook
Executive summary	Dashboard
Root cause analysis	Notebook
SLA reporting	Dashboard
Ad-hoc queries	Notebook

2. Creating Your First Dashboard

Location: the **Dashboards** app → create a dashboard

Dashboard Creation Steps

1. Click "Create dashboard"
2. Name your dashboard (e.g., "Production Overview")
3. Add tiles by clicking "+" or dragging
4. Configure each tile
5. Arrange and resize as needed
6. Save

Dashboard Settings

Setting	Description
Name	Dashboard title
Time frame	Default time range
Segment	Default filter scope
Variables	*"Configure variable filters to monitor different resources within a single dashboard."*
Refresh rate	Auto-refresh interval. *"When you open a dashboard for the first time, the refresh rate is set to Off (no automatic refresh)."* A rate you pick is remembered the next time you open the dashboard
Owner	Dashboard owner
Sharing	Who can view/edit (§6)

> **Sources:** Dashboards (DT docs).

3. Tile and Visualization Types

Tiles

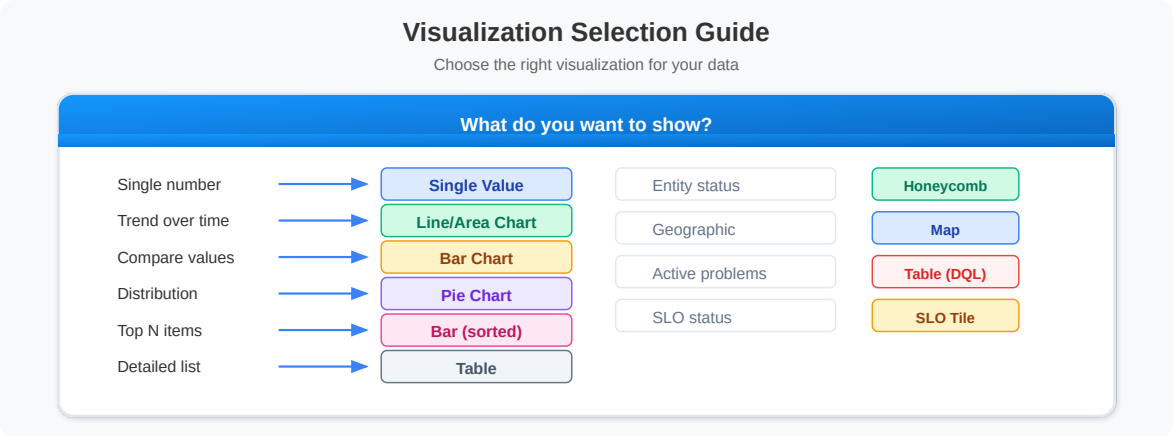
Tile Type	Use Case
Query	A DQL query against Grail, shown with any visualization below
Explore	Point-and-click data exploration, no DQL needed
Code	Data returned by code run as a Dynatrace function
Markdown	Documentation, links
Image	Logos, diagrams
Service-Level Objective	SLO status
Variables	Filters that drive the other tiles

Visualizations for Query and Explore Tiles

Visualization	Use Case
Single value	Key metrics (request count, error rate)
Line / Area / Band chart	Time series data
Bar / Categorical chart	Comparisons and ranked top-N (sort and `limit` in the query)
Table / Record list	Lists and details, including active problems from `dt.davis.problems`
Pie / Donut	Distribution
Honeycomb	Entity health grid
Meter bar / Gauge	Value against a range
Maps (choropleth, dot, connection, bubble)	Geographic data

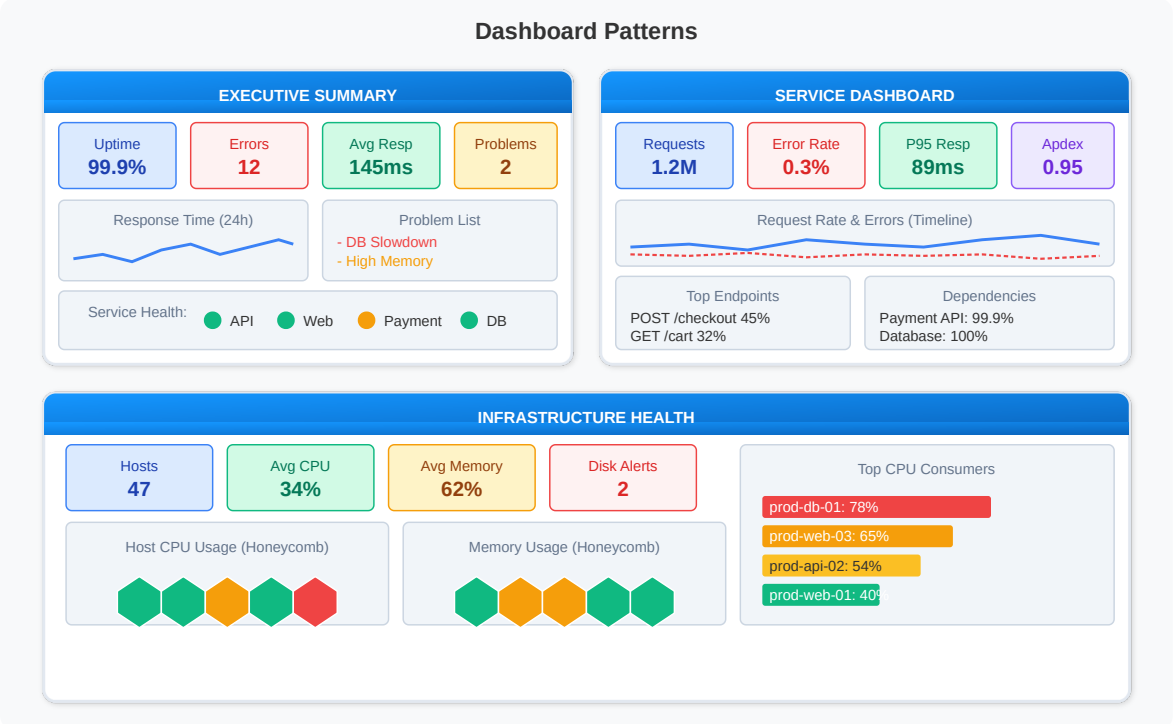
There is no "top list" visualization: a ranked list is a bar or categorical chart (or a table) over a query that ends in `sort … | limit N`.

Visualization Types



> **Sources:** Dashboards (DT docs) — tile types; Edit visualizations (DT docs) — the visualization list.

4. Dashboard Patterns



Pattern: Executive Summary

Key elements:
- **KPI row**: Uptime, Errors, Avg Response, Active Problems
- **Trend chart**: Response time over 24h
- **Problem list**: Current issues
- **Service health**: Status indicators for key services

Pattern: Service Dashboard

Key elements:
- **KPI row**: Requests, Error Rate, P95 Response, Apdex
- **Timeline**: Request rate and errors over time
- **Top endpoints**: Most called API endpoints
- **Dependencies**: Downstream service health

Pattern: Infrastructure Dashboard

Key elements:
- **KPI row**: Host count, Avg CPU, Avg Memory, Disk Alerts
- **Honeycomb views**: Visual host health for CPU and memory
- **Top consumers**: Hosts using most resources

5. Useful Queries for Dashboards

These queries work well as dashboard tiles. Each sets its own timeframe with `from:`, and *"If the timeframe is defined in the query itself, the dropdown list is disabled."* Remove `from:` when the tile should follow the dashboard timeframe.

> **Sources:** Dashboards (DT docs).

Service Health Queries

The service tiles read the service request metrics `dt.service.request.count` and `dt.service.request.failure_count` rather than counting spans. Counting `span.kind == "server"` spans with `span.status_code == "error"` gives a different and misleading

number. A service request is its request root span, which is not always a server span, and *"a request counts as failed only when failure detection marks it as such."* On a validation tenant (1 hour, 10/02/2026) the span version read an error rate of 0.35 % (407 of 115,103 server spans); the service metrics read 4.61 % (4,807 failures in 104,191 requests).

> **Sources:** Service-related concepts (DT docs).

```
// Total request count (Single Value tile)
timeseries requests = sum(dt.service.request.count, scalar: true), from: now()
- 1h
```

```
// Error rate percentage (Single Value tile)
timeseries {
    requests = sum(dt.service.request.count, scalar: true),
    failures = sum(dt.service.request.failure_count, scalar: true)
  }, from: now() - 1h
| fieldsAdd error_rate = round(100.0 * failures / requests, decimals: 2)
```

```
// Top services by request count (Bar chart tile)
timeseries requests = sum(dt.service.request.count, scalar: true), by:
{dt.service.name}, from: now() - 1h
| sort requests desc
| limit 10
```

```
// Service error rates (Table tile)
timeseries {
    requests = sum(dt.service.request.count, scalar: true),
    failures = sum(dt.service.request.failure_count, scalar: true)
  }, by: {dt.service.name}, from: now() - 1h
| fieldsAdd error_rate = round(100.0 * failures / requests, decimals: 2)
| sort error_rate desc
| limit 15
```

Infrastructure Queries

```
// Host count (Single Value tile)
fetch dt.entity.host
| filter state == "RUNNING"
| summarize host_count = count()

// Smartscape note (dt.entity.* is deprecated but still functional): this
query uses the
// classic-only field state, which has NO Smartscape node equivalent
// (Smartscape expresses liveness via node lifetime, not a state field). Keep
the classic
// query above for state detail. For monitoring mode, do not use
monitoringMode — it is
// empty on Kubernetes and Fargate hosts; read billing events instead (ONBRD-
05 § 6).
```

```
// Hosts by OS type (Pie Chart tile)
fetch dt.entity.host
| summarize count = count(), by: {osType}
| sort count desc

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "HOST"
//   | summarize count = count(), by: {os.type}
//   | sort count desc
// Caveat: Smartscape reflects CURRENT live topology and can report fewer
entities
// than the classic entity store; for a pre-migration discovery inventory keep
the
// classic query above.
// Field maps: osType -&gt; os.type (LINUX -&gt; OS_TYPE_LINUX).
```

```
// Host inventory (Table tile)
fetch dt.entity.host
| fields name = entity.name, os = osType, state, cpuCores
| sort name
| limit 25

// Smartscape note (dt.entity.* is deprecated but still functional): this
query uses the
// classic-only field state, which has NO Smartscape node equivalent
// (Smartscape expresses liveness via node lifetime, not a state field). Keep
the classic
// query above for state detail. For monitoring mode, do not use
monitoringMode — it is
// empty on Kubernetes and Fargate hosts; read billing events instead (ONBRD-
05 § 6).
// Other fields do map: osType -&gt; os.type (LINUX -&gt; OS_TYPE_LINUX);
cpuCores -&gt; cores; entity.name -&gt; name.
```

Log Queries

Filter error logs on `status`, not `loglevel`. `loglevel` is the source's own severity, while *"for each log event, a status attribute is created with a value that is a sum of loglevel values"*, and the levels `SEVERE`, `ERROR`, `CRITICAL`, `ALERT`, `FATAL` and `EMERGENCY` all map to `status == "ERROR"`. Java applications log `SEVERE`, so `loglevel == "ERROR"` misses them. On a validation tenant (1 hour, 10/02/2026) `status == "ERROR"` returned 118,086 records and `loglevel == "ERROR"` 63,551; most of the difference was 54,535 `SEVERE` records.

> **Sources:** Automatic log enrichment (DT docs).

```
// Error log count (Single Value tile)
// status groups SEVERE, ERROR, CRITICAL, ALERT, FATAL and EMERGENCY; loglevel
is the raw source value
fetch logs, from: now() - 1h
| filter status == "ERROR"
| summarize error_count = count()
```

```
// Log volume by severity (Pie Chart tile)
fetch logs, from: now() - 1h
| summarize count = count(), by: {loglevel}
| sort count desc
```

```
// Recent errors (Table tile)
fetch logs, from: now() - 1h
| filter status == "ERROR"
| fields timestamp, loglevel, log.source, content
| sort timestamp desc
| limit 10
```

Problem Queries

```
// Active problem count (Single Value tile)
fetch dt.davis.problems, from: now() - 30d
| filter event.status == "ACTIVE"
| summarize active_problems = count()
```

```
// Problems by status (Pie Chart tile)
fetch dt.davis.problems, from: now() - 7d
| summarize problem_count = count(), by: {event.status}
```

```
// Recent problem list (Table tile)
fetch dt.davis.problems, from: now() - 24h
| fields timestamp, display_id, event.name, event.status
| sort timestamp desc
| limit 10
```

6. Sharing and Permissions

Sharing Options

Sharing Level	Who Can Access
Private	Only you, the owner
Specific users or groups	Named users and groups, each with *Can view* or *Can edit*

Sharing Level	Who Can Access
Access for all	*"let everyone in your Dynatrace environment view the document"* (view-only)
Share link	Anyone in the environment who has the link, with the permission chosen when the link was created

A share link can be forwarded: *"anyone in your Dynatrace environment could use it, and they would have the same permissions (Can edit or Can view) that you selected when you created the link."* Use it for view access, not edit.

Segment Filtering

Dashboards can be filtered using segments:
- Set a default segment in dashboard settings
- Viewers can switch segments if they have access
- Data respects viewer's permissions

One Dashboard, Several Views

The Dashboards documentation describes no saved presets. To give one dashboard several views, use **Variables** (viewer-selectable filters such as service, cluster or environment) and a **Default segment**. The documentation's rule of thumb: use segments for filters you reuse across dashboards, and variables *"if you need more control over how a filter is applied"*.

Exporting Dashboards

Format	Use Case
JSON (*Download*)	Backup, migration: *"Download writes the dashboard to a JSON file that you can import"*

> **Edited or generated JSON can stop displaying (SaaS 1.346, staged rollout from 08/25/2026).** *"Starting with Dynatrace version 1.346, Dynatrace applies stricter validation rules to dashboards and won't display dashboards that fail validation until you fix them."* It *"will mainly affect dashboards created or modified via the API or external AI tools."* Import edited or generated JSON into a test environment first and

> confirm it displays (DASH-07). Tenants not yet on 1.346 still display such dashboards.
>
> **Sources:** Share documents (DT docs); Dashboards (DT docs) — segments, variables and Download; SaaS 1.346 (DT docs).

7. Next Steps

Congratulations! You've completed the onboarding series.

What You've Learned

Notebook	Key Topics
ONBRD-01	Tenant access, navigation
ONBRD-02	IAM, SAML, Platform Tokens, parameterized policies
ONBRD-03	ActiveGate deployment
ONBRD-04	Cloud & SaaS integrations
ONBRD-05	OneAgent deployment, primary tags at source
ONBRD-06	Tags, segments, naming conventions, `dt.security_context`
ONBRD-07	Data model, topology
ONBRD-08	DQL fundamentals
ONBRD-09	Workflows alerting
ONBRD-10	Dashboard creation

Where to Go Next — by Topic

Goal	Topic Series
Dashboard strategy + executive reporting	DASH series (8 notebooks)

Goal	Topic Series
Davis AI / anomaly detection / RCA	AIOPS series (8 notebooks)
Workflow automation + AI tasks	WFLOW series (12 notebooks)
Alerting strategy and design	ALERT series (5 notebooks)
Service level objectives	SLO series (6 notebooks)
Deepen DQL — spans / logs / OpenPipeline	SPANS, OPLOGS, OPMIG, OPIPE
Synthetic monitoring	SYNTH series
Web RUM / Mobile RUM	WEBRUM, MOBL
Business events & funnel analysis	BIZEV series
Database monitoring	DBMON series
Cloud integration deep dives	CLOUD series
Kubernetes monitoring	K8S series
OpenTelemetry integration	OTEL series
IAM administration depth	IAM series
Configuration automation / GitOps	AUTOM series
Access control modernization	MZ2POL series (Management Zones → Policies)
Migration playbooks	NRLC + NR2DT (New Relic), SL2DT (Sumo), S2D (Splunk), M2S (Managed → SaaS)
Platform maturity / adoption	ADOPT series
Official training	Dynatrace University
Community	community.dynatrace.com

Dashboard Checklist

- [ ] First dashboard created (using **Dashboards (new)**, not legacy Classic Dashboards)
- [ ] Key metrics visualized
- [ ] Problem overview included
- [ ] SLO / SLI tile evaluated for service-health views
- [ ] Dashboard shared with team
- [ ] Segment filter set

Summary

In this notebook, you learned:

- Difference between Dashboards (new) and Classic Dashboards
- Difference between dashboards and notebooks
- How to create and configure dashboards
- Tile and visualization types and when to use them
- Dashboard patterns for different use cases
- DQL queries that work well in dashboards, including why service tiles read the service metrics and error-log tiles filter on `status`
- That `dt.davis.problems` queries use `event.status` (`ACTIVE` / `CLOSED`), not `status` (`OPEN`)
- How to share dashboards with your team

References

- Dashboards (DT docs)
- Dashboards and notebooks (DT docs)
- Edit visualizations (DT docs)
- Share documents (DT docs)
- Dashboards Classic (DT docs)

- Notebooks (DT docs)
- Dynatrace Query Language (DT docs)
- Dynatrace Community

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